

Trigonometrical Functions and Identities

Choose the most appropriate option (a, b, c or d).

Q 1. If $\tan \theta = a = \frac{1}{4a}$ then $\sec \theta - \tan \theta$ is equal to

- (a) $-2a, \frac{1}{2a}$ (b) $-\frac{1}{2a}, 2a$ (c) $2a$ (d) $\frac{1}{2a}, 2a$

Q 2. $\sec^2 \theta = \frac{4xy}{(x+y)^2}$, where $x \in \mathbb{R}, y \in \mathbb{R}$, is true if and only if

- (a) $x + y \neq 0$ (b) $x = y, x \neq 0$ (c) $x = y$ (d) $x \neq 0, y \neq 0$

Q 3. $\sin^2 \theta = \frac{(x+y)^2}{4xy}$, where $x \in \mathbb{R}$, gives real θ if and only if

- (a) $x + y = 0$ (b) $x = y$ (c) $|x| = |y| \neq 0$ (d) none of these

Q 4. $\operatorname{cosec} \theta = \frac{x^2 - y^2}{x^2 + y^2}$, where $x \in \mathbb{R}, y \in \mathbb{R}$, gives real θ if and only if

- (a) $x = y \neq 0$ (b) $|x| = |y| \neq 0$ (c) $x + y = 0, x \neq 0$ (d) none of these

Q 5. If $\sin \theta + \operatorname{cosec} \theta = 2$ then the value of $\sin^8 \theta + \operatorname{cosec}^8 \theta$ is equal to

- (a) 2 (b) 2^8 (c) 2^4 (d) none of these

Q 6. If $x = r \sin \theta \cdot \cos \phi$, $y = r \sin \theta \cdot \sin \phi$ and $z = r \cos \theta$ then the value of $x^2 + y^2 + z^2$ is independent of

- (a) θ, ϕ (b) r, θ (c) r, ϕ (d) r

Q 7. Let $p = a \cos \theta - b \sin \theta$. Then for all real θ

- (a) $p > \sqrt{a^2 + b^2}$ (b) $p < -\sqrt{a^2 + b^2}$ (c) $-\sqrt{a^2 + b^2} \leq p \leq \sqrt{a^2 + b^2}$ (d) none of these

Q 8. If $0^\circ < \theta < 180^\circ$ then

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots + \sqrt{2(1 + \cos \theta)}}}},$$

there being n number of 2's, is equal to

- (a) $2 \cos \frac{\theta}{2^n}$ (b) $2 \cos \frac{\theta}{2^{n-1}}$ (c) $2 \cos \frac{\theta}{2^{n+1}}$ (d) none of these

Q 9. The value of $\tan \frac{\pi}{16} + 2 \tan \frac{\pi}{8} + 4$ is equal to

- (a) $\cot \frac{\pi}{8}$ (b) $\cot \frac{\pi}{16}$ (c) $\cot \frac{\pi}{16} - 4$ (d) none of these

Q 10. The value of $\sin 78^\circ - \sin 66^\circ - \sin 42^\circ + \sin 6^\circ$ is

- (a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) -1 (d) none of these

Q 11. The value of $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ$ is equal to

- (a) 2 (b) 4 (c) $2 \cdot \frac{\sin 20^\circ}{\sin 40^\circ}$ (d) $4 \cdot \frac{\sin 20^\circ}{\sin 40^\circ}$

Q 12. The maximum value of $1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2\cos\left(\frac{\pi}{4} - \theta\right)$ for real values of θ is

- (a) 3 (b) 5 (c) 4 (d) none of these

Q 13. The minimum value of $\cos 2\theta + \cos \theta$ of real values of θ is

- (a) $-\frac{9}{8}$ (b) 0 (c) -2 (d) none of these

Q 14. The value of $\operatorname{cosec} 10^\circ - \sqrt{3} \sec 10^\circ$ is equal to

- (a) $\frac{1}{2}$ (b) 2 (c) 4 (d) 8

Q 15. The least value of $\cos^2 \theta - 6 \sin \theta \cdot \cos \theta + 3 \sin^2 \theta + 2$ is

- (a) $4 + \sqrt{10}$ (b) $4 - \sqrt{10}$ (c) 0 (d) none of these

Q 16. If $\cos^4 \theta \cdot \sec^2 \alpha$, $\frac{1}{2}$ and $\sin^4 \theta \cdot \operatorname{cosec}^2 \alpha$ are in AP then

$\cos^8 \theta \cdot \cos^6 \alpha$, $\frac{1}{2}$ and $\sin^8 \theta \cdot \operatorname{cosec}^6 \alpha$ are in

- (a) AP (b) GP (c) HP (d) none of these

Q 17. If $\tan \frac{\pi}{9}$, x and $\tan \frac{5\pi}{18}$ are in AP and $\tan \frac{\pi}{9}$, y and $\tan \frac{7\pi}{18}$ are also in AP then

- (a) $2x = y$ (b) $x > y$ (c) $x = y$ (d) none of these

Q 18. If $\cos(x - y)$, $\cos x$ and $\cos(x + y)$ are in HP then $\left| \cos x \cdot \sec \frac{y}{2} \right|$ equals

- (a) 1 (b) 2 (c) $\sqrt{2}$ (d) none of these

- Q 19. If $2 \sin \alpha \cdot \cos \beta \cdot \sin \gamma = \sin \beta \cdot \sin(\alpha + \gamma)$ then $\tan \alpha$, $\tan \beta$ and $\tan \gamma$ are in
 (a) AP (b) GP (c) HP (d) none of these
- Q 20. If $\tan \alpha = \sqrt{a}$, where a is a rational number which is not a perfect square, then which of the following is a rational number ?
 (a) $\sin 2\alpha$ (b) $\tan 2\alpha$ (c) $\cos 2\alpha$ (d) none of these
- Q 21. Let $f(\theta) = \frac{\cot \theta}{1 + \cot \theta}$ and $\alpha + \beta = \frac{5\pi}{4}$. Then the value of $f(\alpha) \cdot f(\beta)$ is
 (a) 2 (b) $-\frac{1}{2}$ (c) $\frac{1}{2}$ (d) none of these
- Q 22. If $\tan \frac{\alpha}{2}$ and $\tan \frac{\beta}{2}$ are the roots of the equation $8x^2 - 26x + 15 = 0$ then $\cos(\alpha + \beta)$ is equal to
 (a) $-\frac{627}{725}$ (b) $\frac{627}{725}$ (c) -1 (d) none of these
- Q 23. If $\sin \alpha + \sin \beta = \cos \alpha - \beta = b$ then $\tan \frac{\alpha - \beta}{2}$ is equal to
 (a) $-\frac{a}{b}$ (b) $-\frac{b}{a}$ (c) $\sqrt{a^2 + b^2}$ (d) none of these
- Q 24. If $0 < \beta < \alpha < \frac{\pi}{4}$, $\cos(\alpha + \beta) = \frac{3}{5}$ and $(\alpha - \beta) = \frac{4}{5}$ then $\sin 2\alpha$ is equal to
 (a) 1 (b) 0 (c) 2 (d) none of these
- Q 25. If $\cos \alpha = \frac{1}{2}\left(x + \frac{1}{x}\right)$, $\cos \beta = \frac{1}{2}\left(y + \frac{1}{y}\right)$ then $\cos(\alpha - \beta)$ is equal to
 (a) $\frac{x}{y} + \frac{y}{x}$ (b) $xy + \frac{1}{xy}$ (c) $\frac{1}{2}\left(\frac{x}{y} + \frac{y}{x}\right)$ (d) none of these
- Q 26. If $\frac{2 \sin \alpha}{1 + \sin \alpha + \cos \alpha} = \lambda$ then $\frac{1 + \sin \alpha - \cos \alpha}{1 + \sin \alpha}$ is equal to
 (a) $\frac{1}{\lambda}$ (b) λ (c) $1 - \lambda$ (d) $1 + \lambda$
- Q 27. If $|\tan A| < 1$, and $|A|$ is acute then $\frac{\sqrt{1 + \sin 2A} + \sqrt{1 - \sin 2A}}{\sqrt{1 + \sin 2A} - \sqrt{1 - \sin 2A}}$ is equal to
 (a) $\tan A$ (b) $-\tan 3\theta$ (c) $\cot A$ (d) $-\cot A$

- Q 28. $\tan \theta \cdot \tan\left(\frac{\pi}{3} + \theta\right) \cdot \tan\left(\frac{\pi}{3} - \theta\right)$ is equal to
- (a) $\tan 2\theta$ (b) $\tan 3\theta$ (c) $\tan^3 \theta$ (d) none of these
- Q 29. The set of all possible values of α in $[-\pi, \pi]$ such that $\sqrt{\frac{1 - \sin \alpha}{1 + \sin \alpha}}$ is equal to $\sec \alpha - \tan \alpha$ is
- (a) $\left[0, \frac{\pi}{2}\right)$ (b) $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$ (c) $[\pi, 0]$ (d) $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$
- Q 30. For all real values of θ , $\cot \theta - 2\cot 2\theta$ is equal to
- (a) $\tan 2\theta$ (b) $\tan \theta$ (c) $-\cot 3\theta$ (d) none of these
- Q 31. Let $a = \cos A + \cos B - \cos(A + B)$ and $b = \sin \frac{A}{2} \sin \frac{B}{2} \cos \frac{A+B}{2}$. Then $a - b$ is equal to
- (a) 1 (b) 0 (c) -1 (d) None of these
- Q 32. If $\tan \theta + \tan\left(\theta + \frac{\pi}{3}\right) + \tan\left(\theta - \frac{\pi}{3}\right) = k \tan 3\theta$ then k is equal to
- (a) 1 (b) 3 (c) $\frac{1}{3}$ (d) none of these
- Q 33. If $a \sec \alpha - c \tan \alpha = d$ and $b \sec \alpha + d \tan \alpha = c$ then
- (a) $a^2 + c^2 = b^2 + d^2$ (b) $a^2 + d^2 = b^2 + c^2$ (c) $a^2 + b^2 = c^2 + d^2$ (d) $ab = cd$
- Q 34. If $\cos 20^\circ - \sin 20^\circ = p$ then $\cos 40^\circ$ is equal to
- (a) $-p\sqrt{2-p^2}$ (b) $p\sqrt{2-p^2}$ (c) $p + \sqrt{2-p^2}$ (d) none of these
- Q 35. If $3\sin \theta + 4\cos \theta = 5$ then the value of $4\sin \theta - 3\cos \theta$ is
- (a) 0 (b) 5 (c) 1 (d) none of these
- Q 36. If $\cos 2x + 2\cos x = 1$ then $\sin^2 x(2 - \cos^2 x)$ is equal to
- (a) 1 (b) -1 (c) $-\sqrt{5}$ (d) $\sqrt{5}$
- Q 37. If $0 < \phi, \frac{\pi}{2}, x = \sum_{n=0}^{\infty} \cos^{2n} \phi, y = \sum_{n=0}^{\infty} \sin^{2n} \phi$ and $z = \sum_{n=0}^{\infty} \cos^{2n} \phi$ then
- (a) $xyz = xz + y$ (b) $xyz = xy + z$ (c) $xyz = x + y + z$ (d) $xyz = yz + x$
- Q 38. Let n be an odd integer. If $\sin n\theta = \sum_{r=0}^n b_r \sum_{r=0}^n b_r \sin^r \theta$ for all real θ then

(a) $b_0 = 1, b_1 = 3$

(b) $b_0 = 0, b_1 = n$

(c) $b_0 = -1, b_1 = n$

(d) $b_0 = 0, b_1 = n^2 - 3n - 3$

Q 39. If $\cos 5\theta = a\cos^5\theta + b\cos^3\theta$ then c is equal to

(a) -5

(b) 1

(c) 5

(d) none of these

Q 40. If $\sin^3 x \cdot \sin 3x = \sum_{m=0}^n c_m \cdot \cos mx$ is an identity in x, where c_m 's are constant then the value of n is

(a) 4

(b) 6

(c) 9

(d) none of these

Q 41. The value of $\sin \frac{\pi}{14} \cdot \sin \frac{3\pi}{14} \cdot \sin \frac{5\pi}{14} \cdot \sin \frac{7\pi}{14} \cdot \sin \frac{9\pi}{14} \cdot \sin \frac{11\pi}{14} \cdot \sin \frac{13\pi}{14}$ is equal to

(a) 1

(b) $\frac{1}{16}$

(c) $\frac{1}{64}$

(d) none of these

Q 42. The numerical value of $\sin \frac{\pi}{18} \cdot \sin \frac{7\pi}{18}$ is equal to

(a) 1

(b) $\frac{1}{8}$

(c) $\frac{1}{4}$

(d) $\frac{1}{2}$

Q 43. The value of $\tan 63^\circ - \cot 63^\circ$ is equal to

(a) $\frac{2}{\sqrt{5}+1} \cdot \sqrt{10-2\sqrt{5}}$

(b) $\frac{2}{\sqrt{5}+1} \cdot \sqrt{10+2\sqrt{5}}$

(c) $\frac{\sqrt{5}-1}{4} \cdot \sqrt{10-2\sqrt{5}}$

(d) none of these

Q 44. The value of $\cos 9^\circ - \sin 9^\circ$ is

(a) $-\frac{\sqrt{5}-\sqrt{5}}{2}$

(b) $\frac{5+\sqrt{5}}{4}$

(c) $\frac{1}{2}\sqrt{5-\sqrt{5}}$

(d) none of these

Q 45. The value of $2\tan \frac{\pi}{10} + 3\sec \frac{\pi}{10} - 4\cos \frac{\pi}{10}$ is

(a) 0

(b) $\sqrt{5}$

(c) 1

(d) none of these

Q 46. The value of $\tan 20^\circ + 2\tan 50^\circ - \tan 70^\circ$ is

(a) 1

(b) 0

(c) $\tan 50^\circ$

(d) none of these

Q 47. If α, β, γ and δ be four angles of a cyclic quadrilateral then the value of $\cos \alpha + \cos \beta + \cos \gamma + \cos \delta$ is

(a) 1

(b) 0

(c) -1

(d) none of these

Q 48. If $4n\alpha = \pi$ then $\cot \alpha \cdot \cot 2\alpha \cdot \cot 3\alpha \dots \cot (2n-1)\alpha$ is equal to

- (a) 1 (b) -1 (c) ∞ (d) none of these

Q 49. The value of $\cos 12^\circ \cdot \cos 24^\circ \cdot \cos 36^\circ \cdot \cos 48^\circ \cdot \cos 72^\circ \cdot \cos 84^\circ$ is

- (a) $\frac{1}{64}$ (b) $\frac{1}{32}$ (c) $\frac{1}{16}$ (d) $\frac{1}{28}$

Q 50. The value of $\cos \frac{\pi}{11} + \cos \frac{3\pi}{11} + \cos \frac{5\pi}{11} + \cos \frac{7\pi}{11} + \cos \frac{9\pi}{11}$ is

- (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) none of these

Q 51. $\sum_{r=1}^{n-1} \cos^2 \frac{r\pi}{n}$ is equal to

- (a) $\frac{n}{2}$ (b) $\frac{n-1}{2}$ (c) $\frac{n}{2} - 1$ (d) none of these

Q 52. The value of $\sin \frac{\pi}{n} + \sin \frac{3\pi}{n} + \sin \frac{5\pi}{n} + \dots$ to n terms is equal to

- (a) 1 (b) 0 (c) $\frac{n}{2}$ (d) none of these

Q 53. The sum of the real roots of $\cos^6 x + \sin^4 x = 1$ in the interval $-\pi < x \leq \pi$ is equal to

- (a) 0 (b) π (c) $-\pi$ (d) none of these

Q 54. The number of real solution of the equation $\cos^7 x + \sin^4 x = 1$ in the interval $[-\pi, \pi]$ is

- (a) 2 (b) 3 (c) 5 (d) none of these

Q 55. If the solutions for θ from the equation $\sin^2 \theta - 2\sin \theta + \lambda = 0$ lie in $\cup \left(2n\pi - \frac{\pi}{6}, 2n\pi + \frac{\pi}{6} \right)$ then the set of possible values of λ is

- (a) $\left[\frac{5}{4}, 1 \right]$ (b) $(-\infty, 1)$ (c) $\left[-\frac{5}{4}, +\infty \right]$ (d) $[1]$

Q 56. If ABCD is a convex quadrilateral such that $4\sec A + 5 = 0$ then the quadratic equation whose roots are $\tan A$ and $\operatorname{cosec} A$ is

- (a) $12x^2 - 29x + 15 = 0$ (b) $12x^2 - 11x - 15 = 0$
(c) $12x^2 + 11x - 15 = 0$ (d) none of these

Q 57. If ABCD is a cyclic quadrilateral such that $12\tan A - 5 = 0$ and $5\cos B + 3 = 0$ then the quadratic equation whose roots are $\cos C$, $\tan D$ is

- (a) $39x^2 - 16x - 48 = 0$ (b) $39x^2 + 88x + 48 = 0$

(c) $39x^2 - 88x + 48 = 0$

(d) none of these

Q 58. The number of real solution of the equation $\sin(e^x) = 2^x + 2^{-x}$ is

(a) 1

(b) 0

(c) 2

(d) infinite

Q 59. The equation $(\cos p - 1)x^2 + (\cos p)x + \sin p = 0$ in x has real roots. Then the set of values of p is

(a) $[0, 2\pi]$

(b) $[-\pi, 0]$

(c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(d) $[0, \pi]$

Q 60. If $e^{\sin x} - e^{-\sin x} - 4 = 0$ then the number of real values of x is

(a) 0

(b) 2

(c) 1

(d) infinite

Q 61. If $\sin \alpha = p$, where $|p| \leq 1$ then the quadratic equation whose roots are $\tan \frac{\alpha}{2}$ and $\cot \frac{\alpha}{2}$ is

(a) $px^2 + 2x + p = 0$

(b) $px^2 - x + p = 0$

(c) $px^2 - 2x + p = 0$

(d) none of these

Q 62. If $\sec \alpha$ and $\operatorname{cosec} \alpha$ are the roots of $x^2 - px + q = 0$ then

(a) $p^2 = q(q - 2)$

(b) $p^2 = q(q + 2)$

(c) $p^2 + q^2 = 2q$

(d) none of these

Q 63. The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is

(a) 0

(b) 5

(c) 6

(d) 10

Q 64. If $x = \alpha, \beta$ satisfies both the equations $\cos^2 x + a \cos x + b = 0$ and $\sin^2 x + p \sin x + q = 0$ then relation between a, b, p and q is

(a) $1 + b + a^2 = p^2 - q - 1$

(b) $a^2 + b^2 = p^2 + q^2$

(c) $2(b + q) = a^2 + p^2 - 2$

(d) none of these

Q 65. If $0 \leq a \leq 3, 0 < b \leq 3$ and the equation $x^2 + 4 + 3 \cos(ax + b) = 2x$ has at least one solution then the value of $a + b$ is

(a) 0

(b) $\frac{\pi}{2}$

(c) π

(d) none of these

Q 66. The equation $\cos \theta = x + \frac{p}{x}$ for all $x \in \mathbb{R}$ has a real solution for θ . Then

(a) $p = \frac{1}{2}$

(b) $p \leq \frac{1}{4}$

(c) $p \geq \frac{1}{4}$

(d) none of these

Q 67. If $f(x) = \frac{\sin 3x}{\sin x}$, where $x \neq n\pi$, then the range of values of $f(x)$ for real values of x is

(a) $[-1, 3]$

(b) $(-\infty, -1)$

(c) $(3, +\infty)$

(d) $[-1, 3)$

Q 68. The set of values of $k \in \mathbb{R}$ such that the equation $\cos 2\theta + \cos \theta + k = 0$ admits of a solution for θ is

- (a) $\left[0, \frac{9}{8}\right]$ (b) $[0, +\infty]$ (c) $[-2, 0]$ (d) none of these

Q 69. The set of values of $\lambda \in \mathbb{R}$ such that $\tan^2\theta + \sec \theta = \lambda$ holds for some θ is

- (a) $(-\infty, 1)$ (b) $(-\infty, -1]$ (c) ϕ (d) $[-1, +\infty)$

Q 70. If $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$ then

- (a) A, B, C must be angles of a triangle
(b) the sum of any two of A, B, C is equal to the third
(c) $A + B + C$ must be an integral multiple of π
(d) none of these

Choose the correct options. One or more options may be correct.

Q 71. If $x = \sin(\alpha - \beta)$, $y = \sin(\beta - \gamma)$, $z = \sin(\gamma - \alpha)$ and

$z = \sin(\gamma - \alpha)$, $\sin(\beta - \delta)$ then

- (a) $x + y + z = 0$ (b) $x + y - z = 0$ (c) $y + z - x = 0$ (d) $x^3 + y^3 + z^3 = 3xyz$

Q 72. $\sin \frac{15\pi}{32} \cdot \sin \frac{7\pi}{16} \cdot \sin \frac{3\pi}{8}$ is equal to

- (a) $\frac{1}{8\sqrt{2} \cos \frac{15\pi}{32}}$ (b) $\frac{1}{8 \sin \frac{\pi}{32}}$ (c) $\frac{1}{4\sqrt{2}} \operatorname{cosec} \frac{\pi}{16}$ (d) $\frac{1}{8\sqrt{2}} \operatorname{cosec} \frac{\pi}{32}$

Q 73. Which of the following is a rational numbers?

- (a) $\sin 15^\circ$ (b) $\cos 15^\circ$ (c) $\sin 15^\circ$ (d) $\sin 15^\circ \cdot \cos 75^\circ$

Q 74. If $a = \frac{1}{5\cos x + 12\sin x}$ then for all real x

- (a) the least positive value of a is $\frac{1}{13}$ (b) the greatest negative value of a is $-\frac{1}{13}$

- (a) $a \leq \frac{1}{13}$ (d) $-\frac{1}{13} \leq a \leq \frac{1}{13}$

Q 75. Let $y = \sin^2 x + \cos^4 x$. Then for all real x

- (a) the maximum value of y is 2 (b) the minimum value of y is $\frac{3}{4}$

$$(c) y \leq \frac{1}{4}$$

$$(d) y \geq -1$$

Q 76. Let $y = \sin x \cdot \sin(60^\circ + x) \cdot \sin(60^\circ - x) \cdot \sin(60^\circ - x)$. Then for all real x

$$(a) \text{ the minimum value of } y \text{ is } -\frac{1}{4}$$

$$(b) \text{ then maximum value of } y \text{ is } 1$$

$$(c) y \leq \frac{1}{4}$$

$$(d) y \geq -1$$

Q 77. Let $f_n(\theta) = \tan \frac{\theta}{2} \cdot (1 + \sec \theta)(1 + \sec 2\theta)(1 + \sec 4\theta) \dots (1 + \sec 2^n \theta)$. Then

$$(a) f_2\left(\frac{\pi}{16}\right) = 1$$

$$(b) f_3\left(\frac{\pi}{32}\right) = 1$$

$$(c) f_4\left(\frac{\pi}{64}\right) = 1$$

$$(d) f_5\left(\frac{\pi}{128}\right) = 1$$

Q 78. Let $0 \leq \theta < \frac{\pi}{2}$ and $x = X \cos \theta + Y \sin \theta$, $y = X \sin \theta - Y \cos \theta$ such that

$$x^2 + 4xy + y^2 = aX^2 + bY^2, \text{ where } a, b \text{ are constants. Then}$$

$$(a) a = -1, b = 3$$

$$(b) \theta = \frac{\pi}{4}$$

$$(c) a = 3, b = -1$$

$$(d) \theta = \frac{\pi}{3}$$

Q 79. If $7 \cos x - 4 \sin x = \lambda \cos(x + \alpha) < \frac{\pi}{2}$, be true for all $x \in \mathbb{R}$ then

$$(a) \lambda = 25$$

$$(b) \alpha = \sin^{-1} \frac{24}{25}$$

$$(c) \lambda = -25$$

$$(d) \alpha = \cos^{-1} \frac{7}{25}$$

Q 80. If $A + B = \frac{\pi}{3}$ and $\cos A + \cos B = 1$ then

$$(a) \cos(A - B) = \frac{1}{3} \quad (b) |\cos A - \cos B| = \sqrt{\frac{2}{3}} \quad (c) (\cos A - B) = -\frac{1}{3} \quad (d) |\cos A - \cos B| = \frac{1}{2\sqrt{3}}$$

Q 81. If $\tan \theta = a \neq 0$, $\tan 2\theta = b \neq 0$ and $\tan \theta + \tan 2\theta = \tan 3\theta$ then

$$(a) a = b$$

$$(b) ab = 1$$

$$(c) a + b = 0$$

$$(d) b = 2a$$

Q 82. $\cos^3 x \cdot \sin 2x = \sum_{m=1}^n a_m \sin mx$ is an identity in x . Then

$$(a) a_3 = \frac{3}{8}, a_2 = 0$$

$$(b) n = 6, a_1 = \frac{1}{2}$$

$$(c) n = 5, a_1 = \frac{1}{4}$$

$$(d) \sum a_m = \frac{3}{4}$$

Q 83. If $1 + \cos(x - y) = 0$ then

$$(a) \cos x - \cos y = 0$$

$$(b) \cos x + \cos y = 0$$

$$(c) \sin x + \sin y = 0$$

$$(d) \cos x + \sin y = 1$$

Q 84. If $A \geq 0$, $B > 0$, $A + B \leq \frac{\pi}{3}$ and $y = \tan A \cdot \tan B$ then

(a) the maximum value of y is 3

(b) the minimum value of y is $\frac{1}{3}$

(c) the maximum value of y is $\frac{1}{3}$

(d) the minimum value of y is 0

Q 85. $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}$ is equal to

(a) an integer

(b) a positive rational number

(c) a negative rational number

(d) an irrational number

Answers

1a	2b	3c	4d	5a	6a	7c	8a	9b	10b
11b	12c	13a	14c	15b	16a	17a	18c	19c	20c
21c	22a	23b	24a	25c	26b	27c	28b	29d	30b
31a	32b	33c	34b	35a	36a	37b	38b	39c	40b
41c	42v	43a	44c	45a	46b	47b	48a	49a	50c
51c	52b	53a	54b	55a	56b	57a	58b	59d	60a
61c	62b	63c	64d	65c	66b	67d	68a	69d	70c
71a,d	72a,d	73c	74a,b	75b,c	76a,c	77a,b,c,d	78b,c	79a,b,d	80b,c
81c	82a,c,d	83b,c	84c,d	85c					

Solutions of Equations and Inequations

Choose the most appropriate option (a, b, c or d).

- Q 1. The number of distinct solutions of $\sin 5\theta \cdot \cos 3\theta = \sin 9\theta \cdot \cos 7\theta$ in $[0, \pi/2]$ is
(a) 4 (b) 5 (c) 8 (d) 9
- Q 2. The number of solutions of $\sin^2 \theta + 3\cos \theta = \sqrt{3}$, $0 \leq \theta \leq 3\pi$, is
(a) 4 (b) 2 (c) 0 (d) None of these
- Q 3. The number of distinct solutions of $\sec \theta + \tan \theta = \sqrt{3}$, $0 \leq \theta \leq 3\pi$, is
(a) 3 (b) 5 (c) 4 (d) 0
- Q 4. The number of solution of the equations $5\sec \theta - 13 = 12\tan \theta$ in $[0, 2\pi]$ is
(a) 2 (b) 1 (c) 4 (d) 0
- Q 5. The equations $p\cos x - q\sin x = r$ admits of a solution for x only if
(a) $r < \max \{p, q\}$ (b) $-\sqrt{p^2 + q^2} < r, \sqrt{p^2 + q^2}$ (c) $r^2 = p^2 + q^2$ (d) none of these
- Q 6. The equations $k\cos x - 2\sin x = k + 1$ is solvable only if k belongs to the interval
(a) $[4, +\infty)$ (b) $[-4, 4]$ (c) $(-\infty, 4]$ (d) none of these
- Q 7. The number of solutions of $\cos \theta + \sqrt{3} \sin \theta = 5$, $0 \leq \theta \leq 5\pi$, is
(a) 4 (b) 0 (c) 5 (d) none of these
- Q 8. If the equation $a_1 + a_2\cos 2x + a_3\sin^2 x = 1$ is satisfied by every real every of x then the number of possible values of the triplet (a_1, a_2, a_3) is
(a) 0 (b) 1 (c) 3 (d) infinite
- Q 9. The number of solutions of the equations $\tan x + \sec x = 2 \cos x$ lying in the interval $[0, 2\pi]$ is
(a) 0 (b) 1 (c) 2 (d) 3
- Q 10. The number of values of x in $[0, 5\pi]$ satisfying the equation $3\cos 2x - 10\cos x + 7 = 0$ is
(a) 5 (b) 6 (c) 8 (d) 10
- Q 11. The smallest positive integral value of p for which the equation $\cos(p\sin x) = \sin(p\cos x)$ in x has solution in $[0, 2\pi]$ is
(a) 2 (b) 1 (c) 3 (d) none of these
- Q 12. If $\sin^2 \theta - 2\sin \theta - 1 = 0$ is to be satisfied for exactly 4 distinct values of $\theta \in [0, n\pi]$, $n \in \mathbb{N}$, then the least value of n is

- (a) 2 (b) 6 (c) 4 (d) 1

Q 13. If $2\tan^2x - 5\sec x$ is equal to 1 for exactly 7 distinct values of $x \in \left[0, \frac{n\pi}{2}\right]$, $n \in \mathbb{N}$, then the greatest value of n is

- (a) 6 (b) 12 (c) 13 (d) 15

Q 14. The number of values of $x \in [0, 2\pi]$ that satisfy $\cot x - \operatorname{cosec} x = 2 \sin x$ is

- (a) 3 (b) 2 (c) 1 (d) 0

Q 15. If $\sin \alpha$, 1 and $\cos 2\alpha$ are in GP then α is equal to

- (a) $n\pi + (-1)^n \frac{\pi}{2}, n \in \mathbb{Z}$ (b) $n\pi + (-1)^{n-1} \frac{\pi}{2}, n \in \mathbb{Z}$ (c) $2n\pi, n \in \mathbb{Z}$ (d) none of these

Q 16. If $\frac{1}{6} \sin \theta$, $\cos \theta$ and $\tan \theta$ are in GP the general value if θ is

- (a) $2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (b) $2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$ (c) $n\pi + (-1)^n \frac{\pi}{3}, n \in \mathbb{Z}$ (d) $n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$

Q 17. The general values of x for which $\cos 2x$, $\frac{1}{2}$ and $\sin 2x$ are in AP are given by

- (a) $n\pi, n\pi \pm \frac{\pi}{2}$ (b) $n\pi, n\pi \pm \frac{\pi}{4}$ (c) $n\pi + \frac{\pi}{4}$ (d) $n\pi$

Q 18. The most general values of θ satisfying $\tan \theta + \tan \left(\frac{3\pi}{4} + \theta\right) =$ are given by

- (a) $n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (b) $2n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$ (c) $2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (d) $n\pi + (-1)^n \frac{\pi}{3}, n \in \mathbb{Z}$

Q 19. The most general solutions of the equation $\sec x = -1 (\sqrt{2} - 1) \tan x$ are given by

- (a) $n\pi + \frac{\pi}{8}$ (b) $2n\pi, 2n\pi \pm \frac{\pi}{4}$ (c) $2n\pi$ (d) none of these

Q 20. The most general solutions of the equation $\sec^2 x = \sqrt{2} (1 - \tan^2 x)$ are given by

- (a) $n\pi + \frac{\pi}{8}$ (b) $n\pi \pm \frac{\pi}{8}$ (c) $n\pi \pm \frac{\pi}{8}$ (d) none of these

Q 21. The most general values of x for which

$$\sin x + \cos x = \min \{1, a^2 - 4a + 6\}$$

are given by

- (a) $2n\pi$ (b) $2n\pi\frac{\pi}{2}$ (c) $n\pi + (-1)^n \cdot \frac{\pi}{4} - \frac{\pi}{4}$ (d) none of these

Q 22. If $\max \{5\sin \theta + 3\sin(\theta - \alpha)\} = 7$ then the set of possible values of α is $\theta \in \mathbb{R}$

- (a) $\left\{x \mid x = 2n\pi \pm \frac{\pi}{3}; n \in \mathbb{Z}\right\}$ (b) $\left\{x \mid x = 2n\pi \pm \frac{2\pi}{3}; n \in \mathbb{Z}\right\}$
 (c) $\left[\frac{\pi}{3}, \frac{2\pi}{3}\right]$ (d) none of these

Q 23. Let α, β be any two positive values of x for which $2\cos x$, $|\cos x|$ and $1-3\cos^2x$ are in GP. The minimum values of $|\alpha - \beta|$ is

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) none of these

Q 24. The sum of all the solution of the equation

$$\cos x \cdot \cos\left(\frac{\pi}{3} + x\right) \cdot \cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}, x \in [0, 6\pi] \text{ is}$$

- (a) 15π (b) 30π (c) $\frac{110\pi}{3}$ (d) none of these

Q 25. The values of $x \in [-2\pi, 2\pi]$ such that $\frac{\sin x + i\cos x}{1+i}, i = \sqrt{-1}$, is purely imaginary, are given by

- (a) $n\pi - \frac{\pi}{4}$ (b) $n\pi + \frac{\pi}{4}$ (c) $n\pi$ (d) none of these

Q 26. If $x \in \left[\frac{5\pi}{2}, \frac{5\pi}{2}\right]$, the greatest positive solution of $1 + \sin^4 x = \cos^2 3x$ is

- (a) π (b) 2π (c) $\frac{5\pi}{2}$ (d) none of these

Q 27. The number of solutions of $|\cos x| = \sin x$, $0 \leq x \leq 4\pi$, is

- (a) 8 (b) 4 (c) 2 (d) none of these

Q 28. The number of solution of $\cos x = 11 + \sin x$, $0 \leq x \leq 3\pi$, is

- (a) 3 (b) 2 (c) 4 (d) none of these

Q 29. Let $[x]$ = the greatest integer less than or equal to x and let $f(x) = \sin x + \cos x$. Then the most

general solutions of $f(x) = \left[f\left(\frac{\pi}{10}\right)\right] f(x) = \left[f\left(\frac{\pi}{10}\right)\right]$ are

- (a) $2n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$ (b) $n\pi, n \in \mathbb{Z}$ (c) $2n\pi, n \in \mathbb{Z}$ (d) none of these

Q 30. The most general solution of $2^{1+|\cos x|+\cos^2 x+|\cos x|^3+\dots\text{to}\infty} = 4$ are given by

- (a) $n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (b) $2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (c) $2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$ (d) none of these

Q 31. If $x \neq \frac{n\pi}{2}$ and $(\cos x)^{\sin^2 x - 3\sin x + 2} = 1$ then all solutions of x are given by

- (a) $2n\pi + \frac{\pi}{2}$ (b) $(2n+1)\pi - \frac{\pi}{2}$ (c) $n\pi + (-1)^n \frac{\pi}{2}$ (d) none of these

Q 32. The solution of the equation $(\sin x + \cos x)^{1+\sin 2x} = 2, -\pi \leq x < \pi$, is

- (a) $\frac{\pi}{2}$ (b) π (c) $\frac{\pi}{4}$ (d) none of these

Q 33. The most general solutions $2^{\sin x} + 2^{\cos x} = 2^{1-1/\sqrt{2}}$ are

- (a) $n\pi - \frac{\pi}{4}$ (b) $n\pi + \frac{\pi}{4}$ (c) $n\pi + (-1)^n \frac{\pi}{4}$ (d) $2n\pi \pm \frac{\pi}{4}$

Q 34. The number of solutions of $16^{\sin^2 x} + 16^{\cos^2 x} = 10, 0 \leq x \leq 2\pi$, is

- (a) 8 (b) 6 (c) 4 (d) 2

Q 35. If $3\sin^2\theta + 2\sin^2\phi = 1$ and $3\sin 2\theta = 2\sin 2\phi, 0 < \theta, \frac{\pi}{2}$ and $0 < \phi < \frac{\pi}{2}$, then the value of $\theta + 2\phi$ is

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) 0 (d) none of these

Q 36. The most general values of θ satisfying the equation $(1 + 2\sin \theta)^2 + (\sqrt{3}\tan \theta - 1)^2 = 0$ are given by

- (a) $n\pi \pm \frac{\pi}{6}$ (b) $n\pi + (-1)^n \frac{7\pi}{6}$ (c) $2n\pi + \frac{7\pi}{6}$ (d) $2n\pi + \frac{11\pi}{6}$

Q 37. If $r > 0, -\pi \leq \theta \leq \pi$ and r, θ satisfy $r\sin \theta = 3$ and $r = 4(1 + \sin \theta)$ then the number of possible solutions of the pair (r, θ) is

- (a) 2 (b) 4 (c) 0 (d) infinite

Q 38. The number of solutions of the equation $x^3 + x^2 + 4x + 2\sin x = 0$ in $0 \leq x \leq 2\pi$ is

- (a) zero (b) one (c) two (d) four

Q 39. The number of real solutions of $\sin e^x \cdot \cos e^x = 2^{x-2} + 2^{-x-2}$ is

- (a) zero (b) one (c) two (d) infinite

Q 40. The least positive nonintegral of $\sin \pi(x^2 + x) - \sin \pi x^2 = 0$ is

- (a) rational
 (b) irrational of the form $\sqrt{p}$
 (c) irrational of the form $\frac{\sqrt{p}-1}{4}$, where p is even integer
 (d) irrational of the form $\frac{\sqrt{p}+1}{4}$, where p is an even integer

Q 41. If the equation $2\cos x + \cos 2\lambda x = 3$ has only one solution then λ is

- (a) 1 (b) a rational number (c) an irrational number (d) none of these

Q 42. If $0 \leq x \leq 2\pi$, $0 \leq y < 2\pi$ and $\sin x + \sin y = 2$ then the value of $x + y$ is

- (a) π (b) $\frac{\pi}{2}$ (c) 3π (d) none of these

Q 43. If $-\pi < x \leq \pi$, $-\pi \leq y \leq \pi$ and $\cos x + \cos y = 2$ then the value $\cos(x - y)$ is

- (a) -1 (b) 0 (c) 1 (d) none of these

Q 44. If $0 \leq x \leq 3\pi$, $0 \leq y \leq 3\pi$ and $\cos x \cdot \sin y = 1$ then the possible number of values of the ordered pair (x, y) is

- (a) 6 (b) 12 (c) 8 (d) 15

Q 45. If $\theta \in [0, 5\pi]$ and $r \in \mathbb{R}$ such that $2\sin \theta = r^4 - 2r^2 + 3$ then maximum number of values of the pair (r, θ) is

- (a) 8 (b) 10 (c) 6 (d) none of these

Q 46. The number of values of x for which $\sin 2x + \cos 4x = 2$ is

- (a) 0 (b) 1 (c) 2 (d) infinite

Q 47. If $2 \sin x + 1 \geq 0$ and $x \in [0, 2\pi]$ then the solution set for x is

- (a) $\left[0, \frac{7\pi}{6}\right]$ (b) $\left[0, \frac{7\pi}{6}\right] \cup \left[\frac{11\pi}{6}, 2\pi\right]$ (c) $\left[\frac{11\pi}{6}, 2\pi\right]$ (d) none of these

Q 48. If $2\cos x < \sqrt{3}$ and $x \in [-\pi, \pi]$ then the solution set for x is

- (a) $\left[-\pi, -\frac{\pi}{6}\right) \cup \left(\frac{\pi}{6}, \pi\right]$ (b) $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$ (c) $\left[-\pi, -\frac{\pi}{6}\right] \cup \left[\frac{\pi}{6}, \pi\right]$ (d) none of these

Q 49. If $\cos x - \sin x \geq 1$ and $0 \leq x \leq 2\pi$ then the solution set for x is

- (a) $\left[0, \frac{\pi}{4}\right] \cup \left[\frac{7\pi}{4}, 2\pi\right]$ (b) $\left[\frac{3\pi}{2}, \frac{7\pi}{4}\right] \cup \{0\}$ (c) $\left[\frac{3\pi}{2}, 2\pi\right] \cup \{0\}$ (d) none of these

Q 50. If $|\tan x| \leq 1$ and $x \in [-\pi, \pi]$ then the solution set for x is

- (a) $\left[-\pi, -\frac{3\pi}{2}\right] \cup \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \pi\right]$ (b) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \pi\right]$ (c) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ (d) none of these

Q 51. If $4\sin^2 x - 8 \sin x + 3 \leq 0$, $0 \leq x \leq 2\pi$, then the solution set for x is

- (a) $\left[0, \frac{\pi}{6}\right]$ (b) $\left[0, \frac{5\pi}{6}\right]$ (c) $\left[\frac{5\pi}{6}, 2\pi\right]$ (d) $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$

Q 52. The set of values of x for which $\sin x \cdot \cos^3 x > \cos x \cdot \sin^3 x$, $0 \leq x \leq 2\pi$, is

- (a) $(0, \pi)$ (b) $\left(0, \frac{\pi}{4}\right)$ (c) $\left(\frac{\pi}{4}, \pi\right)$ (d) none of these

Q 53. The number of values of $x \in [0, 4\pi]$ satisfying is $|\sqrt{3} \cos x - \sin x| \geq 2$

- (a) 2 (b) 0 (c) 4 (d) 8

Type 2

Choose the correct options. One or more options may be correct.

Q 54. If $\alpha \in [-2\pi, 2\pi]$ and $\cos \frac{\alpha}{2} + \sin \frac{\alpha}{2} = \sqrt{2}(\cos 36^\circ - \sin 18^\circ)$ then a value of α is

- (a) $\frac{7\pi}{6}$ (b) $\frac{\pi}{6}$ (c) $-\frac{5\pi}{6}$ (d) $-\frac{\pi}{6}$

Q 55. If $\cos x = \sqrt{1 - \sin 2x}$, $0 \leq x \leq \pi$, then a value of x is

- (a) π (b) 0 (c) $\tan^{-1} 2$ (d) none of these

Q 56. If $\sin 3\theta = \cos 2\theta$ then θ is equal to

- (a) $(4n + 1) \frac{\pi}{2}, (4n + 1) \frac{\pi}{10}$; when n is an even integer only

- (b) $(4n + 1) \frac{\pi}{2}, (4n + 1) \frac{\pi}{10}$; when n is an any integer

(c) $(4n + 1) \frac{\pi}{2}, (4n + 1) \frac{\pi}{10}$; when n is an odd integer only

(d) none of these

Q 57. $\sin \theta + \sqrt{3} \cos \theta = 6x - x^2 - 11, 0 \leq \theta \leq 4\pi, x \in \mathbb{R}$, holds for

(a) no value of x and θ

(b) one value of x and two values of θ

(c) two values of x and two values of θ

(d) two pairs of values of (x, θ)

Q 58. If $0 \leq x \leq 2\pi$ and $|\cos x| \leq \sin x$ then

(a) the set of value of x is $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$

(b) the number of solutions that are integral of $\frac{\pi}{4}$ is three

(c) the sum of the largest and the smallest solution is $\frac{3\pi}{4}$

(d) $x \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right] \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right]$

Q 59. $3^{\sin 2x + 2\cos^2 x} + 3^{1 - \sin 2x + 2\sin^2 x} = 28$ is satisfied by

(a) those values of x for which $\tan x = 1$

(b) those values of x for which $\tan x = -\frac{1}{2}$

(c) those values of x for which $\cos x = 0$

(d) those values of x for which $\tan x = 1$

Q 60. Let $[x]$ = the greatest integer less than or equal to x. The equation $\sin x = [1 + \sin x] + [1 - \cos x]$ has

(a) no solution in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(b) no solution in $\left[\frac{\pi}{2}, \pi\right]$

(c) no solution in $\left[\pi, \frac{3\pi}{2}\right]$

(d) none of these solution for $x \in \mathbb{R}$

Q 61. If $\sin \theta = a$ for exactly one values of $\theta \in \left[0, \frac{7\pi}{3}\right]$ then the value of a is

(a) $\frac{\sqrt{3}}{2}$

(b) 1

(c) 0

(d) -1

Inverse Circular Functions

Choose the most appropriate option (a, b, c or d).

- Q 1. If α satisfies the inequation $x^2 - x - 2 > 0$ then a value exists for
(a) $\sin^{-1} \alpha$ (b) $\sec^{-1} \alpha$ (c) $\cos^{-1} \alpha$ (d) none of these
- Q 2. If α, β are roots of the equation $6x^2 - 11x + 3 = 0$ then
(a) both $\cos^{-1} \alpha$ and $\cos^{-1} \beta$ are real
(b) both $\cos^{-1} \alpha$ and $\cos^{-1} \beta$ are real
(c) both $\cot^{-1} \alpha$ and $\cot^{-1} \alpha$ are real
(d) none of these
- Q 3. Let $f(x) = \sec^{-1} x \tan^{-1} x$. Then $f(x)$ is real for
(a) $x \in [-1, 1]$ (b) $x \in \mathbb{R}$ (c) $x \in (-\infty, -1] \cup [1, \infty)$ (d) none of these
- Q 4. If $\cos^{-1} x - \sin^{-1} x = 0$ then x is equal to
(a) $\pm \frac{1}{\sqrt{2}}$ (b) 1 (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$
- Q 5. $\operatorname{cosec}^{-1}(\cos x)$ is real if
(a) $x \in [-1, 1]$ (b) $x \in \mathbb{R}$
(c) x is an odd multiple of $\frac{\pi}{2}$ (d) x is a multiple of π
- Q 6. The principal value of $\sin \left\{ \sin^{-1} \frac{5\pi}{6} \right\}$ is
(a) $\frac{\pi}{6}$ (b) $\frac{5\pi}{6}$ (c) $\frac{7\pi}{6}$ (d) none of these
- Q 7. The principal value of $\cos^{-1} \left(-\sin \frac{7\pi}{6} \right)$ is
(a) $\frac{5\pi}{3}$ (b) $\frac{7\pi}{6}$ (c) $\frac{\pi}{3}$ (d) none of these
- Q 8. The principal value of $\sin^{-1} \left\{ \cos \left(\sin^{-1} \frac{\sqrt{3}}{2} \right) \right\}$ is
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $-\frac{\pi}{3}$ (d) none of these
- Q 9. The principal value of $\cos^{-1} \left\{ \frac{1}{\sqrt{2}} \left(\frac{9\pi}{10} - \sin \frac{9\pi}{10} \right) \right\}$ is
(a) $\frac{3\pi}{20}$ (b) $-\frac{1}{\sqrt{2}}$ (c) 1 (d) none of these
- Q 10. The value of $\cos \left\{ \tan^{-1} \left(\tan \frac{15\pi}{4} \right) \right\}$ is

- (a) $\frac{1}{\sqrt{2}}$ (b) $-\frac{1}{\sqrt{2}}$ (c) 1 (d) none of these
- Q 11. $-\frac{2\pi}{5}$ is the principal value of
 (a) $\cos^{-1}\left(\cos\frac{7\pi}{5}\right)$ (b) $\sin^{-1}\left(\sin\frac{7\pi}{5}\right)$ (c) $\sec^{-1}\left(\sec\frac{7\pi}{5}\right)$ (d) none of these
- Q 12. If $\cos^{-1}\lambda + \cos^{-1}\mu + \cos^{-1}\nu = 3\pi$ then $\lambda\mu + \mu\nu + \mu\lambda$ is equal to
 (a) -3 (b) 0 (c) 3 (d) -1
- Q 13. If $\sum_{i=1}^{10} \cos^{-1}x_i = 0$ then $\sum_{i=1}^{10} x_i$ is
 (a) 0 (b) 0 (c) 5 (d) none of these
- Q 14. If $\sum_{i=1}^{2n} \sin^{-1}x_i = n\pi$ then $\sum_{i=1}^{2n} x_i$ is equal to
 (a) n (b) 2n (c) $\frac{n(n+1)}{2}$ (d) none of these
- Q 15. The value of $\cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is
 (a) $\frac{\pi}{3}$ (b) 0 (c) $\frac{2\pi}{3}$ (d) none of these
- Q 16. The value of $\tan\left\{2\tan^{-1}\frac{1}{5} - \frac{\pi}{4}\right\}$ is
 (a) 0 (b) 1 (c) $\frac{7}{17}$ (d) none of these
- Q 17. The formula $\cos^{-1}\frac{1-x^2}{1+x^2} = 2\tan^{-1}x$ holds only for
 (a) $x \in \mathbb{R}$ (b) $|x| \leq 1$ (c) $x \in (-1, 1]$ (d) $x \in [1, +\infty)$
- Q 18. $\tan^{-1}a + \tan^{-1}b$, where $a > 0, b > 0, ab > 1$, is equal to
 (a) $\tan^{-1}\frac{a+b}{1-ab}$ (b) $\tan^{-1}\frac{a+b}{1-ab} - \pi$ (c) $\pi + \tan^{-1}\frac{a+b}{1-ab}$ (d) none of these
- Q 19. The set of values of x for which $\tan^{-1}\frac{x}{\sqrt{1-x^2}} = \sin^{-1}x$ hold is
 (a) $\mathbb{R}$ (b) $[-1, 1]$ (c) $[0, 1]$ (d) $[-1, 0]$
- Q 20. $\cos^{-1}\left\{\frac{1}{2}x^2 + \sqrt{1-x^2} \cdot \sqrt{1-\frac{x^2}{4}}\right\} = \cos^{-1}\frac{x}{2}x\cos^{-1}x$ hold is
 (a) $|x| \leq 1$ (b) $x \in \mathbb{R}$ (c) $0 \leq x < 1$ (d) $-1 \leq x < 0$
- Q 21. If $f(x) = \sin^{-1}\left\{\frac{\sqrt{3}}{2}x - \frac{1}{2}\sqrt{1-x^2}\right\}, \frac{1}{2} \leq x \leq 1$, then $f(x)$ is equal to

- (a) $\sin^{-1} \frac{1}{2} - \sin^{-1} x$ (b) $\sin^{-1} x - \frac{\pi}{6}$ (c) $\sin^{-1} x + \frac{\pi}{6}$ (d) none of these

Q 22. The formula $2\sin^{-1} x = \sin^{-1} (2x\sqrt{1-x^2})$ holds for

- (a) $x \in [0, 1]$ (b) $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$ (c) $x \in (-1, 0)$ (d) $x \in \left[-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right]$

Q 23. $\cos^{-1} (\cos x) = x$ is satisfied by

- (a) $x \in \mathbb{R}$ (b) $x \in [0, \pi]$ (c) $x \in [-1, 1]$ (d) none of these

Q 24. If $2\tan^{-1}x + \sin^{-1} \frac{2x}{1+x^2}$ is independent of x then

- (a) $x \in [1, -\infty)$ (b) $x \in [-1, 1]$ (c) $x \in (-\infty, -1)$ (d) none of these

Q 25. If $\tan^{-1}2, \tan^{-1}3$ are angles of a triangle then the third angle is

- (a) $\frac{\pi}{4}$ (b) $\frac{3\pi}{4}$ (c) $\frac{\pi}{2}$ (d) none of these

Q 26. $\sum_{n=1}^3 \tan^{-1} \frac{1}{n}$ is equal to

- (a) 0 (b) π (c) $\frac{\pi}{2}$ (d) none of these

Q 27. The value of $\tan \left\{ \cos^{-1} \left(\frac{4}{5} \right) + \sin^{-1} \left(\frac{2}{\sqrt{13}} \right) \right\}$ is

- (a) $\frac{7}{16}$ (b) $\frac{17}{6}$ (c) $\frac{6}{17}$ (d) none of these

Q 28. If $\sin^{-1} x \cos^{-1} x = \frac{\pi}{6}$ then x is

- (a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) none of these

Q 29. The value of $2\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ is equal to

- (a) $\cot^{-1} x$ (b) $\sec^{-1} x$ (c) $\tan^{-1} x$ (d) none of these

Q 30. The value of $2\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7}$ is

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) π (d) none of these

Q 31. The value of $\cot^{-1}3 + \operatorname{cosec}^{-1} \sqrt{5}$ is

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) none of these

Q 32. $\sin \cot^{-1} \tan \cos^{-1} x$ is equal to

- (a) x (b) $\sqrt{1-x^2}$ (c) $\frac{1}{x}$ (d) none of these
- Q 33. The value of $\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{8}$ is
 (a) π (b) $\frac{\pi}{4}$ (c) $\frac{3\pi}{4}$ (d) none of these
- Q 34. The value of $\tan^2(\sec^{-1}2) + \cot^2(\operatorname{cosec}^{-1}3)$ is
 (a) 13 (b) 15 (c) 11 (d) none of these
- Q 35. $\tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}x\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1}x\right), x \neq 0$, is equal to
 (a) x (b) $2x$ (c) $\frac{2}{x}$ (d) none of these
- Q 36. The number of real solution of the equation is $\sqrt{1+\cos 2x} = \sqrt{2} \sin^{-1}(\sin x), -\pi \leq x \leq \pi$,
 (a) 0 (b) 1 (c) 2 (d) infinite
- Q 37. The number of real solutions of $\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2+x+1} = \frac{\pi}{2}$ is
 (a) zero (b) one (c) two (d) infinite
- Q 38. The number of positive integral solutions of the equation is $\tan^{-1} x + \cos^{-1} \frac{y}{\sqrt{1+y^2}} = \sin^{-1} \frac{3}{\sqrt{10}}$
 (a) one (b) two (c) zero (d) none of these
- Q 39. Considering principal values, the number of solutions of

$$\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$$
 (a) two (b) three (c) one (d) none of these
- Q 40. The number of real solutions of (x, y) , where $|y| = \sin x, y = \cos^{-1}(\cos x), -2\pi \leq x \leq 2\pi$, is
 (a) 2 (b) 1 (c) 3 (d) 4
- Q 41. If $\cos^{-1} x > \sin^{-1} x$ then
 (a) $x < 0$ (b) $-1 < x < 0$ (c) $0 \leq x < \frac{1}{\sqrt{2}}$ (d) $-1 \leq x < \frac{1}{\sqrt{2}}$
- Q 42. If $\cot^{-1} \frac{n}{\pi} > \frac{\pi}{6}, n \in \mathbb{N}$, then the maximum value of n is
 (a) 1 (b) 5 (c) 9 (d) none of these
- Q 43. The set of values of k for which $x^2 - kx + \sin^{-1}(\sin 4) > 0$ for all real x is
 (a) ϕ (b) $(-2, 2)$ (c) $\mathbb{R}$ (d) none of these

Type 2

Choose the correct options. One or more options may be correct.

- Q 44. Let $f(x) = \sin^{-1} x + \cos^{-1} x$. Then $\frac{\pi}{2}$ is equal to
 (a) $f\left(-\frac{1}{2}\right)$ (b) $f(k^2 - 2k + 3)$, k (c) $f\left(\frac{1}{1+k^2}\right)$, $k \in \mathbb{R}$ (d) $f(-2)$
- Q 45. At $x = \frac{3}{2}$, the value is real for
 (a) $\tan^{-1}x$ (b) $\operatorname{cosec}^{-1}x$ (c) $\cos^{-1}2x$ (d) none of these
- Q 46. If $\frac{1}{2} < |x| < 1$ then which of the following are real?
 (a) $\sin^{-1}x$ (b) $\tan^{-1}x$ (c) $\sec^{-1}x$ (d) $\cos^{-1}x$
- Q 47. Let $\tan^{-1}\left(\tan\frac{5\pi}{4}\right) = \alpha$, $\tan^{-1}\left(-\tan\frac{2\pi}{3}\right) = \beta$. Then
 (a) $\alpha < \beta$ (b) $4\alpha - 3\beta = 0$ (c) $\alpha + \beta = \frac{7\pi}{12}$ (d) none of these
- Q 48. Let $f(x) = e^{\cos^{-1}\sin(x+\pi/3)}$. Then
 (a) $f\left(\frac{8\pi}{9}\right) = e^{5\pi/18}$ (b) $f\left(\frac{8\pi}{9}\right) = e^{13\pi/18}$ (c) $f\left(-\frac{7\pi}{4}\right) = e^{\pi/12}$ (d) $f\left(-\frac{7\pi}{4}\right) = e^{11\pi/12}$
- Q 49. If $f(x) = \cos^{-1} x + \cos^{-1} \left\{ \frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2} \right\}$ then
 (a) $f\left(\frac{2}{3}\right) = \frac{\pi}{3}$ (b) $f\left(\frac{2}{3}\right) = 2\cos^{-1}\frac{2}{3} - \frac{\pi}{3}$ (c) $f\left(\frac{1}{3}\right) = \frac{\pi}{3}$ (d) $f\left(\frac{1}{3}\right) = 2\cos^{-1}\frac{1}{3} - \frac{\pi}{3}$
- Q 50. The value of $\tan\left\{\frac{1}{2}\sin^{-1}\frac{2x}{1+x^2} + \frac{1}{2}\cos^{-1}\frac{1-x^2}{1+x^2}\right\}$ is
 (a) $\frac{2x}{1-x^2}$ if $0 \leq x \leq 1$ (b) $\frac{2x}{1-x^2}$ if $x > 1$ (c) not finite if $x > 1$ (d) none of these
- Q 51. α , β and γ are three angles given by
 $\alpha = 2 \tan^{-1}(\sqrt{2} - 1)$, $\beta = \sin^{-1}\frac{1}{\sqrt{2}} + \sin^{-1}\left(-\frac{1}{2}\right)$ and $\gamma = \cos^{-1}\frac{1}{3}$. Then
 (a) $\alpha > \beta$ (b) $\beta - \gamma$ (c) $\alpha < \gamma$ (d) none of these
- Q 52. If $0 < x < 1$ then $\tan^{-1}\frac{\sqrt{1-x^2}}{1+x}$ is equal to
 (a) $\frac{1}{2}\cos^{-1}x$ (b) $\cos^{-1}\sqrt{\frac{1+x}{2}}$ (c) $\sin^{-1}\sqrt{\frac{1-x}{2}}$ (d) none of these
- Q 53. One of the values of x satisfying $\tan(\sec^{-1}x) = \sin \cos^{-1}\frac{1}{\sqrt{5}}$ is

(a) $\frac{\sqrt{5}}{3}$

(b) $\frac{3}{\sqrt{5}}$

(c) $-\frac{\sqrt{5}}{3}$

(d) $-\frac{3}{\sqrt{5}}$

Q 54. If $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$, $\cos^{-1} x - \cos^{-1} y = \frac{\pi}{3}$ then the number of values of (x, y) is

(a) two

(b) four

(c) zero

(d) none of these

Q 55. The solution set of the equation $\cos^{-1} x - \sin^{-1} x = \sin^{-1}(1 - x)$ is

(a) $[-1, 1]$ (b) $\left[0, \frac{1}{2}\right]$ (c) $[-1, 0]$

(d) none of these

Answers

1b	2c	3c	4d	5d	6a	7c	8a	9d	10a
11b	12c	13b	14b	15a	16d	17d	18c	19b	20c
21b	22b	23b	24a	25a	26c	27b	28b	29c	30a
31c	32a	33b	34c	35c	36c	37c	38b	39a	40c
41d	42b	43a	44a,c	45a,b	46a,b,d		47b,c		48b,c
49a,d	50a,c	51b,c		52a,b,c		53b,d	54d		55d

Logarithms

Choose the most appropriate option (a, b, c or d).

- Q 1. If $\log_4 5 = a$ and $\log_5 6 = b$ then $\log_3 2$ is equal to
 (a) $\frac{1}{2a+1}$ (b) $\frac{1}{2b+1}$ (c) $2ab + 1$ (d) $\frac{1}{2ab-1}$
- Q 2. If $\log_{ax} x, \log_{bx} x, \log_{cx} x$ are in HP, where a, b, c, x belong to $(1, +\infty)$, then a, b, c are in
 (a) AP (b) GP (c) HP (d) none of these
- Q 3. If $\log_5 a \cdot \log_a x = 2$ then x is equal to
 (a) 125 (b) a^2 (c) 25 (d) none of these
- Q 4. The value of $3\log \frac{81}{80} + 5\log \frac{25}{24} + 7\log \frac{16}{15}$ is
 (a) $\log 2$ (b) $\log 3$ (c) 1 (d) 0
- Q 5. Let $f(x) = \sqrt{\log_{10} x^2}$. The set of all values of x for which $f(x)$ is real, is
 (a) $[-1, 1]$ (b) $[1, +\infty)$ (c) $(-\infty, -1]$ (d) $(-\infty, -1] \cup [1, +\infty)$
- Q 6. The set of real values of x for which $2^{\log_{\sqrt{2}}(x-1)} > x + 5$ is
 (a) $(-\infty, -1) \cup (4, +\infty)$ (b) $(4, +\infty)$ (c) $(-1, 4)$ (d) none of these
- Q 7. If $a_1, a_2, a_3, \dots$ are positive numbers in GP then $\log a_n, \log a_{n+1}, \log a_{n+2}$ are in
 (a) AP (b) GP (c) HP (d) none of these
- Q 8. If $x = \log_a (bc), y = \log_b (ca)$ and $z = \log_c (ab)$ then which of the following is equal to 1?
 (a) $x + y + z$ (b) $(1+x)^{-1} + (1+y)^{-1} + (1+z)^{-1}$
 (c) xyz (d) none of these
- Q 9. If $x_n > x_{n-1} > \dots > x_2 > x_1 > 1$ then the value of $\log_{x_1} \log_{x_2} \log_{x_3} \dots \log_{x_n} x_n^{x_{n-1}}$ is equal to
 (a) 0 (b) 1 (c) 2 (d) none of these
- Q 10. If $\log x : \log y : \log z = (y - z) : (z - x) : (x - y)$ then
 (a) $x^y \cdot y^z \cdot z^x = 1$ (b) $x^x y^y z^z = 1$ (c) $\sqrt[3]{x} \cdot \sqrt[3]{y} \cdot \sqrt[3]{z} = 1$ (d) none of these
- Q 11. $x^{\log_x a \times \log_a y \times \log_y z}$ is equal to
 (a) x (b) y (c) z (d) none of these
- Q 12. The number of zeros coming immediately after the decimal point in the value of $(0.2)^{25}$ is (given $\log_{10} 2 = 0.30103$)
 (a) 16 (b) 17 (c) 18 (d) none of these
- Q 13. If $[x]$ = the greatest integer less than or equal to x then $[\log_{10} 6730.4]$ has the value
 (a) 6 (b) 4 (c) 5 (d) none of these
- Q 14. The number of solutions of $\log_2 (x + 5) = -6 - x$ is
 (a) 2 (b) 0 (c) 3 (d) none of these

- Q 15. The number of real values of the parameter k for which $(\log_{16} x)^2 - \log_{16} k = 0$ with real coefficients will have exactly one solution is
 (a) 2 (b) 1 (c) 4 (d) none of these
- Q 16. If $\log_{0.5} \sin x = 1 - \log_{0.5} \cos x$ then the number of solutions of $x \in [-2\pi, 2\pi]$ is
 (a) 3 (b) 2 (c) 1 (d) 4
- Q 17. If $\log_{\cos x} \tan x + \log_{\sin x} \cot x = 0$ then the most general solutions of x are
 (a) $n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$ (b) $2n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$ (c) $2n\pi - \frac{3\pi}{4}, n \in \mathbb{Z}$ (d) none of these
- Q 18. The number of values of $x \in [0, n\pi], n \in \mathbb{Z}$, that satisfy $\log_{|\sin|} (1 + \cos x) = 2$, is
 (a) 0 (b) n (c) $2n$ (d) none of these
- Q 19. The value of $\sum_{r=1}^{89} \log_{10} \tan \frac{\pi r}{180}$ is equal to
 (a) 10 (b) 1 (c) 0 (d) none of these
- Q 20. The solution set of $\log_2 |4 - 5x| > 2$ is
 (a) $\left(\frac{8}{5}, +\infty\right)$ (b) $\left(\frac{4}{5}, \frac{8}{5}\right)$ (c) $(-\infty, 0) \cup \left(\frac{8}{5}, +\infty\right)$ (d) none of these
- Q 21. The set of real values of x for which $\log_{0.2} \frac{x+2}{x} \leq 1$ is
 (a) $\left(-\infty, -\frac{5}{2}\right] \cup (0, +\infty)$ (b) $\left[\frac{5}{2}, +\infty\right)$ (c) $(-\infty, -2) \cup (0, +\infty)$ (d) none of these
- Q 22. The set of real values of x satisfying $\log_{1/2} (x^2 - 6x + 15) \geq -2$ is
 (a) $(-\infty, 2]$ (b) $[2, 4]$ (c) $[2, +\infty)$ (d) none of these
- Q 23. If $\log_{0.04} (x - 1) \geq \log_{0.2} (x - 1)$ then x belongs to the interval
 (a) $(1, 2]$ (b) $(-\infty, 2]$ (c) $[2, +\infty)$ (d) none of these
- Q 24. If $\log_{1/\sqrt{2}} \sin x > 0, x \in [0, 4\pi]$, then number of values of x which are integral multiple of $\frac{\pi}{4}$, is
 (a) 6 (b) 12 (c) 3 (d) none of these
- Q 25. If $\log_{\cos x} \sin x \geq 2$ and $0 \leq x \leq 3\pi$ then $\sin x$ lies in the interval
 (a) $\left[\frac{\sqrt{5}-1}{2}, 1\right]$ (b) $\left(0, \frac{\sqrt{5}-1}{2}\right]$ (c) $\left[0, \frac{1}{2}\right]$ (d) none of these
- Q 26. If $\log_{\sqrt{3}} (\sin x + 2\sqrt{2} \cos x) \geq 2, -2\pi \leq x \leq 2\pi$, then the number of solutions of x is
 (a) 0 (b) infinite (c) 3 (d) none of these

Type 2

Choose the correct options. One or more options may be correct.

- Q 27. If $\log_k x \cdot \log_5 k = \log_x 5, k \neq 1, k > 0$, then x is equal to

- (a) k (b) $\frac{1}{5}$ (c) 5 (d) none of these

Q 28. If $x^2 + 4y^2 = 12xy$, $x \in [1, 4]$, $y \in [1, 4]$, then

- (a) the greatest value of $\log_2 (x + 2y)$ is 4
 (b) the least value of $\log_2 (x + 2y)$ is 3
 (c) the range of values of $\log_2 (x + 2y)$ is $[2, 4]$
 (d) the number of integral values of (x, y) is 2 such that $\log_2 (x + 2y)$ is equal to 3

Q 29. If $\log_{0.1} x \leq 2$ then

- (a) then maximum value of x is $\frac{1}{\sqrt{10}}$ (b) x lies between $\frac{1}{100}$ and $\frac{1}{\sqrt{10}}$
 (c) x does not lie between $\frac{1}{100}$ and $\frac{1}{\sqrt{10}}$ (d) the minimum value of x is $\frac{1}{100}$

Q 30. If $x^{3/4(\log_3 x)^2 + \log_3 x - 5/4} = \sqrt{3}$ then x has

- (a) one positive integral value (b) one irrational value
 (c) two positive rational value (d) none of these

Answers

- | | | | | | | | | | |
|-----|-----|-----|-----|-----|-----|-------|---------|----------|---------|
| 1d | 2b | 3c | 4a | 5d | 6a | 7a | 8b | 9b | 10b |
| 11c | 12b | 13d | 14d | 15b | 16b | 17b | 18a | 19c | 20c |
| 21a | 22b | 23c | 24a | 25b | 26d | 27b,c | 28a,c,d | 29a,b,d, | 30a,b,c |

Properties and Solution of Triangles

Type – 1

Choose the most appropriate option (a, b, c or d).

- Q 1. In a $\triangle ABC$, $\frac{c+b}{c-b} \cdot \tan \frac{A}{2}$ is equal to
(a) $\tan\left(\frac{A}{2} + B\right)$ (b) $\cot\left(\frac{A}{2} + B\right)$ (c) $\tan\left(A + \frac{B}{2}\right)$ (d) none of these
- Q 2. If the area of a $\triangle ABC$ be L then $a^2 \sin 2B + b^2 \sin 2A$ is equal to
(a) 2λ (b) λ (c) 4λ (d) none of these
- Q 3. If k be the perimeter of the $\triangle ABC$ then $b \cos^2 \frac{C}{2} + c \cos^2 \frac{B}{2}$ is equal to
(a) k (b) $2k$ (c) $\frac{k}{2}$ (d) none of these
- Q 4. If R denotes circumradius then in $\triangle ABC$, $\frac{b^2 - c^2}{2aR}$ is equal to
(a) $\cos(B - C)$ (b) $\sin(B - C)$ (c) $\cos B - \cos C$ (d) none of these
- Q 5. In a $\triangle ABC$, $\cot \frac{A-B}{2} \cdot \tan \frac{A+B}{2}$ is equal to
(a) $\frac{a+b}{a-b}$ (b) $\frac{a-b}{a+b}$ (c) $\frac{a(a-b)}{b(a+b)}$ (d) none of these
- Q 6. In a $\triangle ABC$, $(c + a + b)(a + b - c) = ab$. The measure of $\angle C$ is
(a) $2b$ (b) $2c$ (c) $3b$ (d) $3a$
- Q 7. In a $\triangle ABC$, $A : B : C = 3 : 5 : 4$. Then $a + b \csc \sqrt{2}$ is equal to
(a) $2b$ (b) $2c$ (c) $3b$ (d) $3a$
- Q 8. The equation $ax^2 + bx + c = 0$, where a, b, c are the sides of a $\triangle ABC$, and the equation $x^2 + \sqrt{2}x + 1 = 0$ have a common root. The measure of $\angle C$ is
(a) 90° (b) 45° (c) 60° (d) none of these
- Q 9. If in a $\triangle ABC$, $a^2 \cos^2 A = b^2 + c^2$ then
(a) $A < \frac{\pi}{4}$ (b) $\frac{\pi}{4} < A < \frac{\pi}{2}$ (c) $A > \frac{\pi}{2}$ (d) $A = \frac{\pi}{2}$
- Q 10. If in a $\triangle ABC$, $\tan \frac{A}{2}$ and $\tan \frac{B}{2}$ satisfy $6x^2 - 5x + 1 = 0$. Then
(a) $a^2 + b^2 > c^2$ (b) $a^2 - b^2 = c^2$ (c) $a^2 + b^2 = c^2$ (d) none of these
- Q 11. Two sides of a triangle are given by the roots of the equation $x^2 - 2\sqrt{3}x + 2 = 0$. The angle between the side is $\pi/3$. The perimeter of the triangle is

- (a) $6 + \sqrt{3}$ (b) $2\sqrt{3} + \sqrt{6}$ (c) $2\sqrt{3} + \sqrt{10}$ (d) none of these
- Q 12. The side of a $\triangle ABC$ are $AB = \sqrt{13}\text{cm}$, $BC = 4\sqrt{3}\text{cm}$ and $CA = 7\text{cm}$. Then $\sin \theta$, where θ is the smallest angle of the triangle, is equal to
- (a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{2}$ (c) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (d) none of these
- Q 13. Two sides of a triangle are $2\sqrt{2}\text{cm}$ and $2\sqrt{3}\text{cm}$ and the angle opposite to the shorter side of the two is $\pi/4$. The largest possible length of the third side is
- (a) $\sqrt{2}(\sqrt{3}+1)\text{cm}$ (b) $(6+\sqrt{2})\text{cm}$ (c) $(\sqrt{6}-\sqrt{2})\text{cm}$ (d) none of these
- Q 14. In a $\triangle ABC$, $a = 8$, $b = 10$ and $c = 12$. The $\angle C$ is equal to
- (a) $\frac{A}{2}$ (b) $2A$ (c) $3A$ (d) none of these
- Q 15. In $\triangle ABC$, if $\tan \frac{A}{2} = \frac{5}{6}$ and $\tan \frac{B}{2} = \frac{20}{37}$ then
- (a) $2a = b + c$ (b) $a > b > c$ (c) $2c = a + b$ (d) none of these
- Q 16. In a $\triangle ABC$, $a = 2b$ and $|A - B| = \frac{\pi}{3}$. The measure of $\angle C$ is
- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) none of these
- Q 17. In a $\triangle ABC$, the tangent of half the difference of two angles is one-third the tangent of half the sum of the two angles. The ratio of sides opposite the angles is
- (a) $2 : 3$ (b) $1 : 3$ (c) $1 : 2$ (d) $3 : 4$
- Q 18. In a $\triangle ABC$, $A = \frac{2\pi}{3}$, $b - c = 3\sqrt{3}\text{cm}$ and $\text{ar}(\triangle ABC) = \frac{9\sqrt{3}}{2}\text{cm}^2$. Then a is
- (a) $6\sqrt{3}\text{cm}$ (b) 9cm (c) 18cm (d) none of these
- Q 19. The area of a $\triangle ABC$ is $a^2 - (b - c)^2$. Then $\tan A$ is equal to
- (a) $\frac{4}{3}$ (b) $\frac{3}{4}$ (c) $\frac{8}{15}$ (d) none of these
- Q 20. In a $\triangle ABC$, $B = 90^\circ$, $AC = h$ and the length of the perpendicular from B to AC is p such that $h = 4p$. If $AB < BC$ then $\angle C$ has the measure
- (a) $\frac{5\pi}{12}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{12}$ (d) none of these
- Q 21. In a $\triangle ABC$, $a = 5$, $b = 4$ and $\tan \frac{C}{2} = \sqrt{\frac{7}{9}}$. The side c is
- (a) 6 (b) 3 (c) 2 (d) none of these
- Q 22. If in a $\triangle ABC$, $AC = 12$, $BC = 13$ and $AB = 5$, then the distance of A from BC is
- (a) $\frac{25}{13}$ (b) $\frac{60}{13}$ (c) $\frac{65}{12}$ (d) none of these

- Q 23. In a $\triangle ABC$, $\cos A = \frac{3}{5}$ and $\cos B = \frac{5}{13}$. The value of $\cos C$ can be
- (a) $\frac{7}{13}$ (b) $\frac{12}{13}$ (c) $\frac{33}{65}$ (d) none of these
- Q 24. In a $\triangle ABC$, $B = \frac{\pi}{8}$ and $C = \frac{5\pi}{8}$. The altitude from A to the side BC is
- (a) $\frac{a}{2}$ (b) $2a$ (c) $\frac{1}{2}(b+c)$ (d) none of these
- Q 25. Two angles of a triangle are $\frac{\pi}{6}$ and $\frac{\pi}{4}$, and the length of the included side is $(\sqrt{3} + 10)$ cm. The area of the triangle is
- (a) $\frac{\sqrt{3}-1}{2}\text{cm}^2$ (b) $\frac{\sqrt{3}}{2}\text{cm}^2$ (c) $\frac{\sqrt{3}+1}{2}\text{cm}^2$ (d) none of these
- Q 26. If in a $\triangle ABC$, $\sin^3 A + \sin^3 B + \sin^3 C = 3\sin A \cdot \sin B \cdot \sin C$, then the value of the determinant
- $$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$
- is
- (a) 0 (b) $(a+b+c)^3$ (c) $(a+b+c)(ab+bc+ca)$ (d) none of these
- Q 27. In a $\triangle ABC$, $A = 90^\circ$. Then $\tan^{-1} \frac{b}{a+c} + \tan^{-1} \frac{c}{a+b}$ is equal to
- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $\tan^{-1} \frac{a}{b+c}$ (d) none of these
- Q 28. If in a $\triangle ABC$, the values of $\cot A$, $\cot B$, $\cot C$ are in AP, then
- (a) a, b, c are in AP (b) a^2, b^2, c^2 are in AP (c) $\cos A, \cos B, \cos C$ are in AP (d) none of these
- Q 29. If in a $\triangle ABC$, $\frac{a}{\cos A} = \frac{b}{\cos B}$, then
- (a) $2\sin A \sin B \sin C = 1$ (b) $\sin^2 A + \sin^2 B = \sin^2 C$ (c) $2\sin A \cos B = \sin C$ (d) none of these
- Q 30. If the sides of a triangle are in GP and the largest angle is twice the smallest angle then the common ratio, which is greater than 1, lies in the interval
- (a) $(1, \sqrt{3})$ (b) $(1, \sqrt{3})$ (c) $\left(1, \frac{\sqrt{5}+1}{2}\right)$ (d) none of these
- Q 31. In a $\triangle ABC$, the sides a, b and c are such that they are the roots of $x^3 - 11x^2 + 38x - 40 = 0$. Then
- $$\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c}$$
- is equal to
- (a) $\frac{3}{4}$ (b) 1 (c) $\frac{9}{16}$ (d) none of these

- Q 32. If $\cos A + \cos B + 2 \cos C = 2$ then the sides of the $\triangle ABC$ are in
 (a) AP (b) GP (c) HP (d) none of these
- Q 33. If in the $\triangle ABC$, the incentre is the middle point of the median AD then $\cos A$ has the value
 (a) $\frac{7}{8}$ (b) $\frac{1}{4}$ (c) $\frac{1}{3}$ (d) $\frac{1}{\sqrt{2}}$
- Q 34. If in a $\triangle ABC$, $3a = b + c$ then $\tan \frac{B}{2} \cdot \tan \frac{C}{2}$ is equal to
 (a) $\tan \frac{A}{2}$ (b) 1 (c) 2 (d) none of these
- Q 35. If in a $\triangle ABC$, $2\cos A \sin C = \sin B$ then the triangle is
 (a) equilateral (b) isosceles (c) right angled (d) none of these
- Q 36. In a $\triangle ABC$, $a = 1$ and the perimeter is six times the AM of the sines of the angle. The measure of $\angle A$ is
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$
- Q 37. In a $\triangle ABC$, $\angle A > \angle B$. If $\sin A$ and $\sin B$ satisfy the equation $3\sin x - 4\sin^3 x - k = 0$, $0 < k < 1$, then $\angle C$ is
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{6}$
- Q 38. In a $\triangle ABC$ the side a , b and c are in AP. Then $\left(\tan \frac{A}{2} + \tan \frac{C}{2} \right) : \cot \frac{B}{2}$ is
 (a) 3 : 2 (b) 1 : 2 (c) 3 : 4 (d) none of these
- Q 39. If the sides of a triangle are proportional to the cosines of the opposite angles then the triangle is
 (a) right angled (b) equilateral (c) obtuse angled (d) none of these
- Q 40. If in a $\triangle ABC$, $\cos^2 \frac{A}{2} + a \cos^2 \frac{C}{2} = \frac{3b}{2}$, then a , b , c are in
 (a) GP (b) HP (c) AP (d) none of these
- Q 41. The side of a triangle are in AP and its area is $\frac{3}{5}$ x (area of an equilateral triangle of the same perimeter). Then the ratio of the sides of
 (a) 1 : 2 : 3 (b) 3 : 5 : 7 (c) 3 : 4 (d) none of these
- Q 42. If BD, BE and CF are the medians of a $\triangle ABC$ then
 (a) 1 : 2 : 3 (b) 3 : 5 : 7 (c) 1 : 3 : 5 (d) none of these
- Q 43. $\sin A$, $\sin B$, $\sin C$ are in AP for the $\triangle ABC$. Then
 (a) the altitudes are in AP (b) the altitudes are in HP
 (c) The medians are in GP (d) the medians are in AP

Q 44. AD is median of the $\triangle ABC$. If AE and AF are medians of the triangles ABD and ADC respectively, and $AD = m_1$, $AE = m_2$, $AF = m_3$, then $\frac{a^2}{8}$ is

- (a) $m_2^2 + m_3^2 - 2m_1^2$ (b) $m_1^2 + m_2^2 - 2m_3^2$ (c) $m_2^2 + m_3^2 - m_1^2$ (d) none of these

Q 45. The ratio of the distance of the orthocenter of an acute-angled $\triangle ABC$ from the sides BC, AC and AB is

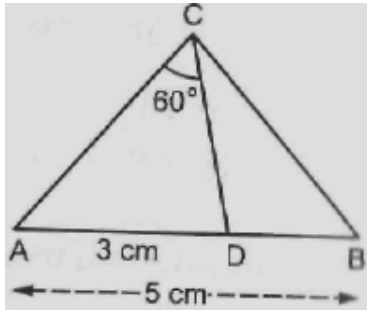
- (a) $\cos A : \cos B : \cos C$ (b) $\sin A : \sin B : \sin C$ (c) $\sec A : \sec B : \sec C$ (d) none of these

Q 46. In a $\triangle ABC$, I is the incentre. The ratio $IA : IB : IC$ is equal to

- (a) $\operatorname{cosec} \frac{A}{2} : \operatorname{cosec} \frac{B}{2} : \operatorname{cosec} \frac{C}{2}$ (b) $\sin \frac{A}{2} : \sin \frac{B}{2} : \sin \frac{C}{2}$
(c) $\sec \frac{A}{2} : \sec \frac{B}{2} : \sec \frac{C}{2}$ (d) none of these

Q 47. In the figure, ABC is a triangle in which $C = 90^\circ$ and $AB = 5$ cm.

D is a point on AB such that $AD = 3$ cm and $\angle ACD = 60^\circ$. Then the length of AC is



- (a) $5\sqrt{\frac{3}{7}}$ cm (b) $\sqrt{\frac{7}{3}}$ cm (c) $\frac{3}{\sqrt{7}}$ cm (d) none of these

Q 48. If α, β, γ are the altitudes of $\triangle ABC$ and $2s$ denotes its perimeter then $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ is equal to

- (a) $\frac{\Delta}{s}$ (b) $\frac{s}{\Delta}$ (c) $s \cdot \Delta$ (d) none of these

Q 49. In a $\triangle ABC$, the sides are in the ratio 4 : 5 : 6. The ratio of the circumradius and the inradius is

- (a) 8 : 7 (b) 3 : 2 (c) 7 : 3 (d) 16 : 7

Q 50. In a $\triangle ABC$, R = circumradius and r = inradius. The value of $\frac{a \cos A + b \cos B + c \cos C}{a + b + c}$ is equal to

- (a) $\frac{R}{r}$ (b) $\frac{R}{2r}$ (c) $\frac{r}{R}$ (d) $\frac{2r}{R}$

Q 51. In a $\triangle ABC$, $2s$ = perimeter and R = circumradius. Then s/R is equal to

- (a) $\sin A + \sin B + \sin C$ (b) $\cos A + \cos B + \cos C$
(c) $\sin \frac{A}{2} + \sin \frac{B}{2} + \sin \frac{C}{2}$ (d) none of these

Q 52. The ratio of the circumradius and inradius of an equilateral triangle is

- (a) 3 : 1 (b) 1 : 1 (c) $2 : \sqrt{3}$ (d) 2 : 1

Q 53. If in a $\triangle ABC$, $a^2 + b^2 + c^2 = 8R^2$, where R = circumradius, then the triangle is

- (a) equilateral (b) isosceles (c) right angled (d) none of these

- Q 54. In an equilateral triangle, (circumradius): (inradius): (exradius) is equal to
 (a) 1 : 1 : 1 (b) 1 : 2 : 3 (c) 2 : 1 : 3 (d) 3 : 2 : 4
- Q 55. The diameter of the circumcircle of a triangle with sides 5 cm, 6 cm and 7 cm is
 (a) $\frac{3\sqrt{6}}{2}$ cm (b) $2\sqrt{6}$ cm (c) $\frac{35}{48}$ cm (d) none of these
- Q 56. If in a triangle, R and r are the circumradius and inradius respectively then the HM of the exradii of the triangle is
 (a) 3r (b) 2R (c) R + r (d) none of these
- Q 57. The angles of a right-triangle are in AP. The ratio of the inradius and the perimeter is
 (a) $(2 - \sqrt{3}) : 2\sqrt{3}$ (b) $1 : 8\sqrt{3}(2 + \sqrt{3})$ (c) $(2 + \sqrt{3}) : 4\sqrt{3}$ (d) none of these
- Q 58. A $\triangle ABC$ is right angled at B. Then the diameter of the incircle of the triangle is
 (a) $2(c + a - b)$ (b) $c + a - 2b$ (c) $c + a - b$ (d) none of these
- Q 59. If the exradii of a triangle are in HP then corresponding sides are in
 (a) AP (b) GP (c) HP (d) none of these
- Q 60. In a $\triangle ABC$, the inradius and three exradii are r, r_1 , r_2 and r_3 respectively. In usual notations the value of $r : r_1, r_2, r_3$ is equal to
 (a) 2Δ (b) Δ^2 (c) $\frac{abc}{4R}$ (d) none of these
- Q 61. In a $\triangle ABC$, the perimeter = 2s and the exradii are r_1 , r_2 and r_3 . Then $r_1r_2 + r_2r_3 + r_3r_1$ is equal to
 (a) s^2 (b) $2s^2$ (c) $3s^2$ (d) $4s^2$
- Q 62. In a $\triangle ABC$, $\tan A \cdot \tan B \cdot \tan C = 9$. For such triangles, if then $\tan^2 A + \tan^2 B + \tan^2 C = k$
 (a) $9\sqrt[3]{3} < k < 27$ (b) $k \not\leq 27$ (c) $k < 9\sqrt[3]{3}$ (d) $k \not\leq 27$
- Q 63. If the $\triangle ABC$ is acute angled at C then
 (a) $\cos 2A + \cos 2B - \cos 2C < 1$ (b) $\cos 2A + \cos 2B + \cos 2C < 1$
 (c) $\cos^2 A + \cos^2 B + \cos^2 C < 1$ (d) none of these
- Q 64. If for a $\triangle ABC$, $\cot A \cdot \cot B \cdot \cot C > 0$ then the triangle is
 (a) right angled (b) acute angled (c) obtuse angled (d) all these options are possible
- Q 65. In a $\triangle ABC$, $\cos B \cdot \cos C + \sin B \cdot \sin C \cdot \sin^2 A = 1$. Then the triangle is
 (a) right-angled isosceles (b) isosceles whose equal angle are greater than $\pi/4$
 (c) equilateral (d) none of these
- Q 66. In a $\triangle ABC$, the angles A and B are two values of θ satisfying $\sqrt{3} \cos \theta + \sin \theta = k$, $k|k| < 2$. The triangle
 (a) is acute angled (b) is right angled (c) is obtuse angled (d) has one angle = $\pi/3$
- Q 67. If in an obtuse-angled triangle the obtuse angle is $3\pi/4$ and the other two angles are equal to two values of θ satisfying $a \tan \theta + b \sec \theta = c$, where $|b| \leq \sqrt{a^2 + c^2}$, then $a^2 - c^2$ is equal to
 (a) ac (b) 2ac (c) a/c (d) none of these

- Q 68. Let A_0, A_1, A_2, A_3, A_4 and A_5 be the consecutive vertices of a regular hexagon inscribed in a unit circle. The product of the lengths of A_0A_1, A_0A_2 and A_0A_4 is
- (a) $\frac{3}{4}$ (b) $3\sqrt{3}$ (c) 3 (d) $\frac{3\sqrt{3}}{2}$
- Q 69. The area of a circle is A_1 and the area of a regular pentagon inscribed in the circle is A_2 . Then $A_1 : A_2$ is
- (a) $\frac{\pi}{5} \cos \frac{\pi}{10}$ (b) $\frac{2\pi}{5} \sec \frac{\pi}{10}$ (c) $\frac{2\pi}{5} \operatorname{cosec} \frac{\pi}{10}$ (d) none of these
- Q 70. The area of a cyclic quadrilateral ABCD is $(3\sqrt{3})/4$. The radius of the circle circumscribing it is 1. If $AB = 1, BD = \sqrt{3}$ then BC. CD is equal to
- (a) 2 (b) $3 - \frac{1}{\sqrt{3}}$ (c) $3\sqrt{3} + 1$ (d) none of these

Type 2

Choose the correct options. One or more options may be correct.

- Q 71. If in $\triangle ABC$, $a = 6, b = 3$ and $\cos(A - B) = \frac{4}{5}$ then
- (a) $C = \frac{\pi}{4}$ (b) $A = \sin^{-1} \frac{2}{\sqrt{5}}$ (c) $\operatorname{ar}(\triangle ABC) = 9$ (d) none of these
- Q 72. The number of possible triangles ABC in which $BC = \sqrt{11}$ cm, $CA = \sqrt{13}$ cm and $A = 60^\circ$ is
- (a) 0 (b) 1 (c) 2 (d) none of these
- Q 73. In a $\triangle ABC$, $A = \frac{\pi}{3}$ and $b : c = 2 : 3$. If $\tan \alpha = \frac{\sqrt{3}}{5}, 0 < \alpha = \frac{\pi}{2}$, then
- (a) $B = 60^\circ + \alpha$ (b) $C = 60^\circ + \alpha$ (c) $B = 60^\circ - \alpha$ (d) $C = 60^\circ - \alpha$
- Q 74. In a triangle the cosines of two angles are inversely proportional to the sides opposite the angles. The triangle is
- (a) isosceles (b) equilateral (c) right angled (d) none of these
- Q 75. In a $\triangle ABC$, the line segments AD, BE and CF are three altitudes. If R is the circumradius of the $\triangle ABC$, a side of the $\triangle DEF$ will be
- (a) $R \sin 2A$ (b) $c \cos B$ (c) $a \sin A$ (d) $b \cos B$
- Q 76. In a $\triangle ABC$, $\tan A$ and $\tan B$ are the roots of the equation $ab(x^2 + 1) = c^2x$, where a, b and c are the sides of the triangle. Then
- (a) $\tan(A - B) = \frac{a^2 - b^2}{2ab}$ (b) $\cot C = 0$ (c) $\sin^2 A + \sin^2 B = 1$ (d) none of these
- Q 77. The distances of the circumcentre of the acute-angled $\triangle ABC$ from the sides BC, CA and AB are in the ratio
- (a) $a \sin A : b \sin B : c \sin C$ (b) $\cos A : \cos B : \cos C$
(c) $a \cot A : b \cot B : c \cot C$ (d) none of these
- Q 78. In any $\triangle ABC$, $\sum a^3 \sin(B - C)$ is equal to
- (a) $\sum a(\sin B - \sin C)$ (b) $\sum a^2(\cos^2 B - \cos^2 C)$ (c) 0 (d) none of these
- Q 79. In $\triangle ABC$, $2 \cos \frac{A - C}{2} = \frac{a + c}{\sqrt{a^2 + c^2 - ac}}$. Then
- (a) $B = \frac{\pi}{3}$ (b) $B = C$ (c) A, B, C are in AP (d) $B + C = A$

- Q 80. Let $L \sin \theta = 10 + \log \sin \theta$. The number of triangles ABC such that $\log b + 10 = \log c + L \sin B$ is
 (a) one (b) two (c) infinite (d) none of these
- Q 81. In any triangle ABC, $\sin \frac{A}{2}$ is
 (a) less than $\frac{a}{b+c}$ (b) less than or equal to $\frac{b+c}{a}$
 (c) greater than $\frac{2a}{a+b+c}$ (d) none of these
- Q 82. In a $\triangle ABC$, $\tan C < 0$. Then
 (a) $\tan A \cdot \tan B < 1$ (b) $\tan A \tan B > 1$
 (c) $\tan A + \tan B + \tan C < 0$ (d) $\tan A + \tan B + \tan C > 0$
- Q 83. In a $\triangle ABC$, $\tan A$ and $\tan B$ satisfy the inequation $3x^2 - 4x + 3 < 0$ Then
 (a) $a^2 + b^2 + ab > c^2$ (b) $a^2 + b^2 - ab < c^2$ (c) $a^2 + b^2 > c^2$ (d) none of these
- Q 84. In a $\triangle ABC$, $\cos A + \cos B + \cos C > 1$ only if the triangle is
 (a) acute angled
 (b) obtuse angled
 (c) right angled
 (d) the nature of the triangle cannot be determined

Answers

1a	2c	3c	4b	5a	6c	7c	8b	9c	10c
11b	12b	13a	14b	15b	16b	17c	18b	19c	20c
21a	22b	23c	24a	25c	26a	27a	28b	29c	30b
31c	32a	33b	34d	35b	36c	37c	38d	39b	50c
41b	42c	43b	44a	45c	46a	47a	48b	49d	50c
51a	52d	53c	54c	55d	56a	57a	58c	59a	60b
61a	62b	63a	64b	65a	66c	67b	68c	69b	70a
71b,c	72c	73b,c	74a,c	75a,d	76a,b,c		77b,c	78a,b,c	79a,c 80a
81b	82a,c	83a,b	84d						

Heights and Distances

Type – 1

Choose the most appropriate option (a, b, c or d).

- Q 1. A 6-ft tall man finds that the angle of elevation of the top of a 24-ft-high pillar and the angle of depression of its base are complementary angles.
(a) $2\sqrt{3}$ ft (b) $8\sqrt{3}$ ft (C) $6\sqrt{3}$ ft (d) none of these
- Q 2. A rocket of height h metres is fired vertically upwards. Its velocity at time t seconds is $(2t + 3)$ metres/second. If the angle of elevation of the top of the rocket from a point on the ground after 1 second of firing is $\pi/6$ and after 3 seconds it is $\pi/3$ then the distance of the point from the rocket is
(a) $14\sqrt{3}$ metres (b) $7\sqrt{3}$ metres
(c) $2\sqrt{3}$ metres (d) cannot be found without the value of h
- Q 3. Three vertical poles of heights h_1 , h_2 and h_3 at the vertices A, B and of a $\triangle ABC$ subtend angles α, β and γ respectively at the circumcentre of the triangle. If $\cot \alpha$, $\cot \beta$ and $\cot \gamma$ are in AP then h_1 , h_2 , h_3 are in
(a) AP (b) GP (c) HP (d) none of these
- Q 4. The angle of elevation of the top of a hill from each of the vertices A, B, C of a horizontal triangle is α . The height of the hill is
(a) $b \tan \alpha \cdot \operatorname{Cosec} B$ (b) $\frac{1}{2} a \tan \alpha \cdot \operatorname{Cosec} A$ (c) $\frac{1}{2} c \tan \alpha \cdot \operatorname{Cosec} C$ (d) none of these
- Q 5. A piece of paper in the shape of a sector of a circle of radius 10 cm and of angle 216° just covers the lateral surface of a right circular cone of vertical angle 2θ . Then $\sin \theta$ is
(a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{3}{4}$ (d) none of these
- Q 6. The angle of elevation of the top of a vertical pole when observed from each vertex of a regular hexagon is $\pi/3$. If the area of the circle circumscribing the hexagon be A metre² then the area of the hexagon is
(a) $\frac{3\sqrt{3}}{8} A$ metre² (b) $\frac{\sqrt{3}}{\pi} A$ metre² (c) $\frac{3\sqrt{3}}{4\pi} A$ metre² (d) $\frac{3\sqrt{3}}{2\pi} A$ metre²
- Q 7. A vertical pole PO is standing at the centre O of a square ABCD. If AC subtends an angle 90° at the top, P, of the pole then the angle subtended by a side of the square at P is
(a) 45° (b) 30° (c) 60° (d) none of these
- Q 8. A vertical lamp-post of height 9 metres stands at the corner of a rectangular field. The angle of elevation of its from the farthest corner is 30° , while from another corner it is 45° . The area of the field is
(a) $81\sqrt{2}$ metre² (b) $9\sqrt{2}$ metre² (c) $81\sqrt{3}$ metre² (d) $9\sqrt{3}$ metre²

- Q 9. A vertical lamp-post, 6 m high, stand at a distance of 2 m from a wall, 4 m high. A 1.5-m-tall man starts to walk away from the wall on the other side of the wall, in line with the lamp-post. The maximum distance to which the man can walk remaining in the shadow is
- (a) $\frac{5}{2}$ m (b) $\frac{3}{2}$ m (c) 4m (d) none of these
- Q 10. A circular of radius 3 cm is suspended horizontally from a point 4 cm vertically above the centre by 4 strings attached at equal intervals to its circumference. If the angle between two consecutive strings be θ then $\cos \theta$ is
- (a) $\frac{4}{5}$ (b) $\frac{4}{25}$ (c) $\frac{4}{25}$ (d) none of these

Type 2

Choose the correct options. One or more options may be correct.

- Q 11. A man standing between two vertical posts finds that the angle subtended at his eyes by the tops of the post is a right angle. If the heights of the two posts are two times and four times the height of the man, and the distance between them is equal to the length of the longer post, then the ratio of the distance of the man from the shorter and the longer post is
- (a) 3 : 1 (b) 2 : 3 (c) 3 : 2 (d) 1 : 3
- Q 12. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar, is
- (a) $\sqrt{3} : 1$ (b) 1 : 3 (c) $1 : \sqrt{3}$ (d) $\sqrt{3} : 2$

Answers

1c 2b 3c 4b 5a 6d 7c 8a 9a 10c
 11a,d 12c